Event Analysis and Event-Use Case Table

1. Perfect Technology Assumption

All identified events and use cases were reviewed to check that each one is a genuine business requirement. Acceptance criteria that only handle technology failure or

system performance were not turned into events (table below). Technical and security functions are also not included: login or logout, change password, and backup or

restore database. E25 (device status review) is kept only as a read-only query because it is a stated stakeholder need (US-18).

Excluded item Source Reason	Excluded item Source Reason	Excluded item Source Reason
Hub or sensor offline or faulty US-18; AC 18.4, 18.5 Handles a device fault, not a business activity	Hub or sensor offline or faulty US-18; AC 18.4, 18.5 Handles a device fault, not a business activity	Hub or sensor offline or faulty US-18; AC 18.4, 18.5 Handles a device fault, not a business activity
Hub low-battery alert and power outage US-18; AC 18.6 Handles a power and hardware problem	Hub low-battery alert and power outage US-18; AC 18.6 Handles a power and hardware problem	Hub low-battery alert and power outage US-18; AC 18.6 Handles a power and hardware problem
Sensor events delivered within 5 seconds US-18; AC 18.7 Performance requirement, not a business event	Sensor events delivered within 5 seconds US-18; AC 18.7 Performance requirement, not a business event	Sensor events delivered within 5 seconds US-18; AC 18.7 Performance requirement, not a business event
Emergency call to 123 not connected US-19; AC 19.5 Handles a communication failure	Emergency call to 123 not connected US-19; AC 19.5 Handles a communication failure	Emergency call to 123 not connected US-19; AC 19.5 Handles a communication failure
Voice-message delivery status (Failed, Delivered, etc.) US-12; AC 12.2 Handles a delivery failure	Voice-message delivery status (Failed, Delivered, etc.) US-12; AC 12.2 Handles a delivery failure	Voice-message delivery status (Failed, Delivered, etc.) US-12; AC 12.2 Handles a delivery failure

2. External events

Event ID	Event description	Actor	Corresponding use case
E01	Routine information is set or changed	Care Coordinator	Manage routine profile
E02	Member gives informed consent	Care Coordinator (with Elderly Member)	Record informed consent
E03	Six-month consent review is completed	Care Coordinator	Complete consent review
E04	Member withdraws consent	Elderly Member	Withdraw consent
E05	Care visit is requested	FamilyContact, Caregiver Team Lead	Book care visit
E06	Caregiver is assigned to a visit	Caregiver Team Lead	Assign caregiver to visit
E07	Caregiver arrives and checks in	Field Caregiver	Check in to visit
E08	Caregiver leaves and checks out	Field Caregiver	Check out of visit
E09	Caregiver submits visit note	Field Caregiver	Submit visit note
E10	Family rates a completed visit	Family Contact	Rate completed visit
E11	Family subscribes to a care plan	Family Contact	Subscribe to care plan
E12	Family opens daily summary	Family Contact	View daily wellbeing summary
E13	Family sends a voice message	Family Contact	Send voice message
E14	Member answers the speaker check-in	Elderly Member	Confirm safety

Event ID	Event description	Actor	Corresponding use case
E15	Member pauses monitoring	Elderly Member	Pause monitoring
E16	Member enables Away Mode	Elderly Member	Enable Away Mode
E17	Member disables Away Mode	Elderly Member	Disable Away Mode
E18	Family contact answers escalation call	Family Contact	Record family response
E19	Dispatched caregiver arrives at member's home	Field Caregiver	Record urgent-visit arrival
E20	Volunteer responds to an alert	Neighbor Volunteer	Respond to non-urgent alert
E21	Emergency Services confirm dispatch	Emergency Services (123)	Record emergency dispatch outcome
E22	Director requests service performance report	Founder and Director	Review service performance
E23	Director reviews subscriptions and invoices	Founder and Director	Review subscriptions and invoices
E24	Director reviews caregiver records and payroll	Founder and Director	Review caregiver records and payroll
E25	Technology Partner requests device status (query only under PTA)	Technology Partner	Review device and sensor status

3. Temporal events

Event ID	Event description	Time / schedule	Corresponding use case
E26	Medication reminder time reached	Daily, at each reminder_time saved in the member's medication schedule.	Play medication reminder
E27	Morning routine cutoff passes	Daily, at the member's morning cutoff (derived from wake-up time). Exact cutoff (e.g. 11:30 or 12:00) still to be confirmed with the Care Coordinator. Checked against kitchen-sensor state.	Check morning inactivity
E28	24-hour advance-notice point reached	Once per scheduled visit, at visit scheduled_time minus 24 hours.	Send advance visit notice
E29	End of day (daily summary time)	Daily, once per member at a fixed end-of-day time (exact time to be confirmed). Display is later adjusted to the family contact's time zone.	Generate daily wellbeing report
E30	End of month (monthly report time)	Monthly, once per member; the report presents a three-month activity trend.	Generate monthly wellbeing report
E31	Six-month consent review date reached	Once per consent record, at review_date (consent_date plus 6 months).	Flag consent review as due
E32	Consent expiry date reached without review	Once per consent record, at expiry_date, if the review was not completed.	Expire consent and stop monitoring
E33	Monitoring pause expires	Once per pause, at pause_expiry_time (no later than 24 hours after the pause began).	Resume monitoring after pause
E34	Subscription month begins	Monthly, on each monthly anniversary of an Active subscription's start_date.	Generate monthly invoice
E35	Pay period ends	At the end of each pay period (length to be confirmed), per caregiver.	Calculate caregiver payroll
E36	Background-check expiry date reached	Annually, per caregiver, at bg_check_expiry_date.	Flag expired background check
E37	Speaker check-in response window elapses	5 minutes after the Arabic safety prompt, if no confirmation was received.	Escalate alert to family contacts
E38	Family-unreachable window elapses	10 minutes of family contacts remaining unreachable (the report does not state whether this is counted from the alert or from the first call; AC 13.3 implies	Dispatch nearest available caregiver

4. State events

Event ID	Event description	Entity whose state changes	State change / triggering condition	Corresponding use case
E39	Hard fall detected	Member (via fall detector)	Fall status changes from normal to hard fall detected while monitoring is active (not suppressed by Away Mode).	Dispatch emergency services (123) and notify family
E40	Unusual non-fall activity detected	Home activity (sensor event)	Sensor activity changes from matching the member's routine to deviating from it (e.g. door opened at night), while monitoring is active and Away Mode is off.	Start speaker safety check-in
E41	Fall detector unworn or charging in daytime	Fall detector	Wear status changes from Worn to Unworn or Charging during daytime hours.	Send fall detector wear reminder
E42	Non-urgent alert raised	Alert	A new alert is created with a non-urgent type, and verified volunteers exist in the member's district.	Notify neighbor volunteers

5. Event-Use Case Table

Event ID	Event Description	Event Type	Triggering Actor / Condition	Corresponding Use Case
E01	Routine information is set or changed	External	Care Coordinator	Manage routine profile
E02	Member gives informed consent	External	Care Coordinator (with Elderly Member)	Record informed consent
E03	Six-month consent review is completed	External	Care Coordinator	Complete consent review
E04	Member withdraws consent	External	Elderly Member	Withdraw consent
E05	Care visit is requested	External	Family Contact; Caregiver Team Lead	Book care visit
E06	Caregiver is assigned to a visit	External	Caregiver Team Lead	Assign caregiver to visit
E07	Caregiver arrives and checks in	External	Field Caregiver	Check in to visit
E08	Caregiver leaves and checks out	External	Field Caregiver	Check out of visit
E09	Caregiver submits visit note	External	Field Caregiver	Submit visit note
E10	Family rates a completed visit	External	Family Contact	Rate completed visit
E11	Family subscribes to a care plan	External	Family Contact	Subscribe to care plan
E12	Family opens daily summary	External	Family Contact	View daily wellbeing summary
E13	Family sends a voice message	External	Family Contact	Send voice message
E14	Member answers the speaker check-in	External	Elderly Member	Confirm safety
E15	Member pauses monitoring	External	Elderly Member	Pause monitoring
E16	Member enables Away Mode	External	Elderly Member	Enable Away Mode

E17	Member disables Away Mode	External	Elderly Member	Disable Away Mode
E18	Family contact answers escalation call	External	Family Contact	Record family response
E19	Dispatched caregiver arrives at member's home	External	Field Caregiver	Record urgent-visit arrival
E20	Volunteer responds to an alert	External	Neighbor Volunteer	Respond to non-urgent alert
E21	Emergency Services confirm dispatch	External	Emergency Services (123)	Record emergency dispatch outcome
E22	Director requests service performance report	External	Founder and Director	Review service performance
E23	Director reviews subscriptions and invoices	External	Founder and Director	Review subscriptions and invoices
E24	Director reviews caregiver records and payroll	External	Founder and Director	Review caregiver records and payroll
E25	Technology Partner requests device status (query only under PTA)	External	Technology Partner	Review device and sensor status
E26	Medication reminder time reached	Temporal	Daily, at each reminder_time saved in the member's medication schedule.	Play medication reminder
E27	Morning routine cutoff passes	Temporal	Daily, at the member's morning cutoff (derived from wake-up time). Exact cutoff (e.g. 11:30 or 12:00) still to be confirmed with the Care Coordinator. Checked against kitchen-sensor state.	Check morning inactivity
E28	24-hour advance-notice point reached	Temporal	Once per scheduled visit, at visit scheduled_time minus 24 hours.	Send advance visit notice
E29	End of day (daily summary time)	Temporal	Daily, once per member at a fixed end-of-day time (exact time to be confirmed). Display is later adjusted to the family contact's time zone.	Generate daily wellbeing report
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E31	Six-month consent review date reached	Temporal	Once per consent record, at review_date (consent_date plus 6 months).	Flag consent review as due

E32	Consent expiry date reached without review	Temporal	Once per consent record, at expiry_date, if the review was not completed.	Expire consent and stop monitoring
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E34	Subscription month begins	Temporal	Monthly, on each monthly anniversary of an Active subscription's start_date.	Generate monthly invoice
E35	Pay period ends	Temporal	At the end of each pay period (length to be confirmed), per caregiver.	Calculate caregiver payroll
E36	Background-check expiry date reached	Temporal	Annually, per caregiver, at bg_check_expiry_date.	Flag expired background check
E37	Speaker check-in response window elapses	Temporal	5 minutes after the Arabic safety prompt, if no confirmation was received.	Escalate alert to family contacts
E38	Family-unreachable window elapses	Temporal	10 minutes of family contacts remaining unreachable (the report does not state whether this is counted from the alert or from the first call; AC 13.3 implies from the family calls).	Dispatch nearest available caregiver
E39	Hard fall detected	State	Fall status changes from normal to hard fall detected while monitoring is active (not suppressed by Away Mode).	Dispatch emergency services (123) and notify family
E40	Unusual non-fall activity detected	State	Sensor activity changes from matching the member's routine to deviating from it (e.g. door opened at night), while monitoring is active and Away Mode is off.	Start speaker safety check-in
E41	Fall detector unworn or charging in daytime	State	Wear status changes from Worn to Unworn or Charging during daytime hours.	Send fall detector wear reminder
E42	Non-urgent alert raised	State	A new alert is created with a non-urgent type, and verified volunteers exist in the member's district.	Notify neighbor volunteers